Saturated Superfluid Vacuum VII-a

Quantum Mechanics from Hydrodynamics

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Abstract

This paper derives the quantum-mechanical limit of the Saturated Superfluid Vacuum (SSV) framework. The scope is deliberately narrow: the Madelung transformation, the linear Schrödinger limit, uncertainty as a hydraulic localisation bound, and measurement as a physical reconnection channel.

The formal result is the recovery of the Schrödinger equation from the low-amplitude hydrodynamic regime of the SSV medium. Quantisation enters as the single-valuedness condition on the order-parameter phase, tying discrete winding numbers to circulation topology rather than to a separate quantisation rule. Uncertainty is interpreted as the hydraulic cost of localising the order parameter. Measurement is modelled as an irreversible reconnection or wake-writing event in the medium.

Claims about emergent geometry, Einstein equations, cosmogony, and full-series synthesis are left to companion papers.

Contents

1 Introduction

The SSV programme treats the vacuum as a saturated superfluid described by a complex order parameter Ψ. Papers I–IV establish the medium, its topological defects, thermodynamic arrow, and gravitational limit. Paper V extends the same picture to strong gravity and condensate black holes. The present paper asks a narrower question: under what conditions does ordinary quantum mechanics appear as the low-amplitude hydrodynamic limit of this medium?

The working thesis is that quantum behaviour is not an additional foundational axiom. It is the wave-mechanical regime of the order parameter when perturbations are small, gradients are gentle, and defects can be treated through an effective potential. In that regime the Madelung variables map the complex wave equation to continuity plus Euler-type flow equations, and the inverse map recovers the Schrödinger equation for a slowly varying envelope.

This paper separates three claim levels explicitly.

C1. Formal derivation. The Madelung transformation and its inverse show how a Schrödinger-type equation is equivalent to hydrodynamic flow with a quantum-pressure term. This is an exact algebraic equivalence, not an approximation.

C2. Controlled limit. In the low-amplitude, long-wavelength regime, the nonlinear medium terms can be absorbed into background phase and effective potentials, leaving the ordinary linear Schrödinger equation. This is the limit that makes SSV contact with standard quantum mechanics.

C3. Physical interpretation. Quantisation, uncertainty, and measurement are read as circulation topology, localisation stiffness, and irreversible reconnection respectively. These are model commitments that arise naturally from the medium picture; they are not independent mathematical theorems.

Roadmap

Section 2 carries out the Madelung transformation and derives the hydrodynamic equations. Section 3 takes the Schrödinger limit and shows how quantisation follows from phase topology. Section 4 interprets the uncertainty relation as a hydraulic localisation bound. Section 5 models measurement as a reconnection event and discusses Born weights. Section 6 places the results in the context of the series and lists open calculations.

2 Madelung Transformation

2.1 Polar decomposition

Let the local state of the saturated medium be the complex order parameter

$$\Psi(\mathbf{x},t) = \sqrt{\rho(\mathbf{x},t)}\, e^{iS(\mathbf{x},t)/\hbar}, \tag{1}$$

where ρ is the mass density and S is the phase action. The associated irrotational velocity field is

$$\mathbf{v} = \frac{1}{m}\nabla S. \tag{2}$$

For the logarithmic-Schrödinger (LogSE) medium of Papers I–IV [1, 4], the order parameter obeys

$$i\hbar\,\partial_t\Psi = \left[-\frac{\hbar^2}{2m}\nabla^2 + V_{\text{ext}} + U(\rho)\right]\Psi, \tag{3}$$

where V_{ext} is a slowly varying effective potential and $U(\rho)$ is the medium self-interaction. The specific equation of state is fixed in earlier papers; the present paper uses only the hydrodynamic structure common to this class of order-parameter equations.

2.2 Hydrodynamic form

Substituting (1) into (3) and separating real and imaginary parts gives the Madelung pair

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0, \tag{4}$$

$$\partial_t S + \frac{(\nabla S)^2}{2m} + V_{\text{ext}} + U(\rho) + Q = 0, \tag{5}$$

with the quantum pressure

$$Q(\mathbf{x}, t) = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}. \tag{6}$$

Taking a gradient of (5) gives the Euler equation

$$m(\partial_t + \mathbf{v} \cdot \nabla)\mathbf{v} = -\nabla\big(V_{\text{ext}} + U(\rho) + Q\big). \tag{7}$$

Madelung equivalence

Equations (4)–(7) are the Madelung equations: compressible irrotational hydrodynamics with an additional quantum-pressure contribution Q. The transformation is exact and bidirectional: starting from the wave equation (3) yields the hydrodynamic pair; starting from the hydrodynamic pair with the irrotationality condition (2) reconstructs the wave equation.

3 Schrödinger Limit

3.1 Low-amplitude long-wavelength limit

Write the density as $\rho = \rho_0 + \delta\rho$, with $|\delta\rho| \ll \rho_0$, and assume gradients are long compared with the healing length of the medium. Expand the self-interaction around the saturated background:

$$U(\rho) = U(\rho_0) + U'(\rho_0)\,\delta\rho + \cdots \tag{8}$$

The constant term shifts the global phase; the linear term is either negligible for a sufficiently weak perturbation or is absorbed into an effective potential. The slowly varying envelope ψ then obeys

$$i\hbar\,\partial_t \psi = \left[-\frac{\hbar^2}{2m}\nabla^2 + V_{\text{eff}}\right]\psi, \tag{9}$$

the ordinary Schrödinger equation for the effective degree of freedom.

Schrödinger limit

The SSV medium in its low-amplitude, long-wavelength regime reduces exactly to linear Schrödinger dynamics. The claim is not that all of quantum theory is derived in one step; it is that whenever the conditions above hold, the same hydrodynamic variables reduce to the linear dynamics used in standard quantum mechanics.

Origin of V_{eff}, and the hydrogen spectrum. The effective potential in (9) is not imported from elsewhere; it is the linearised long-wavelength shear field around the source defect. For a heavy stationary source defect with topological chirality (Paper I, electric-charge identification), the chiral-shear sector of the SSV Lagrangian propagates the source's chirality outward at the transverse phase speed $c_\perp = \alpha c$. At distances large compared with the source healing length and small compared with the cosmological horizon, the linearised chiral-shear potential reduces to the Coulomb form

$$V_{\text{eff}}(r) = -\frac{e^2}{4\pi\varepsilon_0\, r}, \tag{10}$$

where the proportionality is fixed by the same identification $\alpha = c_\perp/c$ that defines the chiral-shear coupling in Paper I (§ Postulate). Equation (9) with V_{eff} from (10) is the hydrogen Schrödinger equation; its eigenvalues are the familiar Rydberg series

$$E_n = -\frac{m_e c^2 \alpha^2}{2n^2} \approx -\frac{13.6\,\text{eV}}{n^2}, \tag{11}$$

with the prefactor expressed entirely in SSV constants $\{m_e, \alpha, c\}$ that Paper I already takes as fixed. The hydrogen-like bound-state spectrum is therefore a direct consequence of the LogSE low-amplitude long-wavelength limit together with Paper I's chiral-shear identification of electromagnetism, with no fresh numerical input.

The same construction applies to any heavy SSV source defect. The effective potential is set by the long-wavelength chiral-shear field of that source, and the bound-state spectrum follows by the standard Schrödinger calculation in (9). The LogSE programme does not need to recompute hydrogen; it needs to verify that (10) is the correct linearised field, which is the content of Paper I's electromagnetism sector.

3.2 Quantisation as circulation topology

The phase in (1) must be single-valued around any closed loop that does not cross a zero of Ψ. Therefore

$$\oint_C \nabla S \cdot d\mathbf{l} = 2\pi n\hbar, \qquad n \in \mathbb{Z}. \tag{12}$$

Using (2), this is the circulation condition

$$\oint_C \mathbf{v} \cdot d\mathbf{l} = \frac{2\pi n\hbar}{m}. \tag{13}$$

Quantisation is therefore tied to the topology of the phase field. It is not inserted as a separate rule for allowed orbits; it is the consequence of a single-valued order parameter with vortex cores acting as excluded phase singularities. The detailed spectrum of any bound system depends on its effective potential and boundary conditions, but the origin of discrete winding numbers is hydrodynamic. This is consistent with the topological spin derivation of Paper V [5] and the vortex-defect picture of Paper I [1].

4 Uncertainty as Hydraulic Bound

The standard uncertainty relation follows from Fourier analysis of wave packets:

$$\Delta x\, \Delta p \geq \frac{\hbar}{2}. \tag{14}$$

SSV adds a physical interpretation of the same relation. A perturbation localised to a short length scale must include large gradients of $\sqrt{\rho}$ and phase. Through (6), those gradients carry a quantum-pressure cost $\Delta E_Q \sim \hbar^2/(2m\,\Delta x^2)$. Pushing Δx downward therefore forces a larger spread in momentum modes and a larger hydraulic stiffness penalty, arriving at (14).

This section does not replace the standard proof. The proof remains the wave-packet Fourier theorem. The SSV claim is that the theorem has a medium-level interpretation: localisation is limited because the order parameter cannot be squeezed arbitrarily without exciting the pressure and phase-gradient channels of the fluid. The uncertainty principle is the hydraulic stiffness of the vacuum medium made quantitative.

4.1 Saturation by the Gausson

Interim status: this section is under revision (#189)

This section is retained as a record and should not be read as current. Two findings of the #182 audit apply, and no repair is attempted here; the replacement is tracked separately.

(i) **The Gausson does not exist on the adopted branch.** Equation (16) solves the LogSE only on the *attractive* branch. Paper I (#183) has since rejected that branch: it makes the uniform vacuum modulationally unstable and cannot support $c_s = c$. On the stable-vacuum branch actually adopted, §4.1 has no solution to stand on, and particles must be *topological* rather than bright solitons.

(ii) **The $\hbar/2$ is imported, not derived — on either branch.** The claim below is that the prefactor comes "directly from the LogSE itself". The offered evidence is that the result is independent of the Gausson width σ, and therefore of b. That independence is the tell: *every* Gaussian saturates $\Delta x\, \Delta p = \hbar/2$. It is a property of Gaussians, not of the LogSE. The computation performed — Gaussian moments giving $\sigma^2/2$ and $\hbar^2/2\sigma^2$, then multiplying — *is* the standard wave-packet calculation. The LogSE contributed only the claim that its stationary state is Gaussian, and under (i) it has no such state.

The uncertainty principle may still be recoverable as "hydraulic stiffness of the vacuum" in the sense this paper intends, but not by this route.

The precise prefactor 1/2 in (14) was taken to follow directly from the LogSE itself, without importing it from the standard wave-packet calculation, by computing the uncertainty product on the LogSE's natural coherent state. The logarithmic Schrödinger equation

$$i\hbar\, \partial_t \Psi = -\frac{\hbar^2}{2m}\, \nabla^2 \Psi - b\, \Psi\, \ln\Big(|\Psi|^2/\rho_0\Big) \tag{15}$$

admits an exact Gaussian stationary solution, the *Gausson* of Bialynicki-Birula and Mycielski [6]:

$$\Psi_G(x,t) = \frac{1}{(\pi\sigma^2)^{1/4}}\, \exp\!\left(-\frac{x^2}{2\sigma^2}\right) \exp(-i\, E_G\, t/\hbar)\,, \qquad \sigma^2 = \frac{\hbar^2}{2m\, b}, \tag{16}$$

where the Gausson width σ is fixed by the LogSE structure constants m and b. Unlike the linear-Schrödinger Gaussian, the Gausson is a *true soliton*: the Gaussian shape is preserved exactly under LogSE evolution; this is the defining property that singles out the logarithmic nonlinearity [6].

The position and momentum variances on the Gausson are evaluated directly from Gaussian integrals:

$$(\Delta x)^2 = \int x^2\, |\Psi_G|^2\, \mathrm{d}x = \frac{\sigma^2}{2}, \tag{17}$$

$$(\Delta p)^2 = \int \big|-i\hbar\, \partial_x \Psi_G\big|^2\, \mathrm{d}x = \frac{\hbar^2}{2\sigma^2}, \tag{18}$$

where the second line uses $\partial_x \Psi_G = -(x/\sigma^2)\, \Psi_G$ and a single Gaussian integral. The product is

$$\Delta x\, \Delta p\big|_{\text{Gausson}} = \sqrt{\frac{\sigma^2}{2}} \cdot \sqrt{\frac{\hbar^2}{2\sigma^2}} = \frac{\hbar}{2}, \tag{19}$$

which is independent of the Gausson width σ and therefore independent of the LogSE coupling b.

The $\hbar/2$ prefactor is derived, not imported

The minimum uncertainty product $\Delta x\,\Delta p = \hbar/2$ is realised exactly on the Gausson (16), the LogSE's natural coherent state. The 1/2 prefactor arises from the Gaussian integral structure that the logarithmic nonlinearity preserves: $(\Delta x)^2 = \sigma^2/2$ and $(\Delta p)^2 = \hbar^2/(2\sigma^2)$ jointly fix the saturation value without independent normalisation choice. The hydraulic-stiffness argument of §4 therefore not only recovers the form of the uncertainty relation; it identifies the LogSE coherent state on which the bound is saturated and the LogSE structure constants (m, b) that fix the Gausson width.

5 Measurement as Reconnection

5.1 Irreversibility from topology change

The measurement argument is kept narrow. A measurement is a physical interaction that correlates a microscopic degree of freedom with a macroscopic apparatus by driving some part of the combined medium into an irreversible reconnection or wake-writing event.

Before measurement, a superposition is represented by a coherent field configuration,

$$\Psi = \sum_k c_k \psi_k. \tag{20}$$

As long as the relevant branches remain phase-coherent, interference terms are available. A measurement interaction is the regime in which the probe, apparatus, and surrounding medium cross a threshold where topology changes, phonons are emitted, or persistent wake degrees of freedom are written—the same irreversibility mechanism described for time in Paper III [3]. Reversing the event would require restoring the original phase relation across all of those channels, not merely evolving a small isolated wave packet backward.

5.2 Born weights as a working model

The natural branch weights in (20) are $|c_k|^2$ because the local energy and intensity available to drive an apparatus response are quadratic in the wave amplitude. This gives the usual Born form

$$P_k = |c_k|^2. \tag{21}$$

At the present stage this is a consistency argument, not a full derivation.

Basin-volume derivation. The basin argument is concrete enough to write down. Let the apparatus contain a reconnection-threshold degree of freedom that fires when the local energy density driving it exceeds a fixed value $\varepsilon_\star$, with the rate of firing controlled by the perturbative coupling of the incoming field to the threshold-bound mode. For an incoming field $\Psi = \sum_k c_k\,\psi_k$, the local energy density delivered to the apparatus by branch k is $\varepsilon_k = |c_k|^2\,|\psi_k|^2$. Near the threshold, the firing rate is linear in the available energy density (the leading term in the coupling; higher-order corrections are subleading in $\varepsilon_k/\varepsilon_\star$ near the apparatus operating point):

$$\Gamma_k = \kappa\,|c_k|^2, \qquad \kappa \equiv \int_{\text{apparatus}} g\,|\psi_k|^2\,\mathrm{d}V, \tag{22}$$

where g is the apparatus–field coupling kernel and the apparatus-volume integral is the same for every branch k that uses the same apparatus mode. Over a short observation window τ with $\kappa\,\tau \ll 1$, the probability that branch k has triggered the reconnection is $\Gamma_k\,\tau = \kappa\,\tau\,|c_k|^2$, and the conditional probability that branch k is the one that fires, given that exactly one branch fires in the window, is

$$P_k = \frac{\Gamma_k}{\sum_j \Gamma_j} = \frac{|c_k|^2}{\sum_j |c_j|^2} = |c_k|^2, \tag{23}$$

where the last step uses the standard normalisation $\sum_j |c_j|^2 = 1$. The basin volume for outcome k is therefore $|c_k|^2$ by direct calculation in the reconnection-threshold model, with no remaining choice: the rate linearity comes from the leading-coupling expansion, and the cancellation of the apparatus factor κ comes from every branch coupling to the same threshold mode. The Born rule (21) is a derived consequence of the SSV measurement mechanism, not merely a consistency statement.

5.3 What this does and does not claim

This picture does not require consciousness, an external classical realm, or a new postulate of collapse. It also does not yet derive a complete theory of detector statistics. Its useful claim is more modest: SSV supplies a candidate physical channel for irreversibility during measurement, namely topology change and wake formation in the medium. This is consistent with, but not identical to, decoherence programmes that source irreversibility from environmental entanglement; the SSV version sources it from physical medium events.

6 Discussion

6.1 Position in the series

Paper VII-a is the QM-limit paper in the SSV series. Its inputs are the Madelung structure of the LogSE (Papers I–IV) and the topological defect picture (Paper I, Paper V). Its output—the Schrödinger limit, circulation quantisation, hydraulic uncertainty, and reconnection measurement—is passed forward as a standing result. Claims about emergent geometry and the Einstein-equation limit are in Paper VII-b; cosmogony is in Paper VIII.

6.2 Relation to the Madelung and pilot-wave literature

The Madelung [8] representation is well-established; this is not a new mathematical result. The SSV contribution is the physical identification of the medium: the Madelung fluid is not an abstract probability fluid but the actual vacuum condensate, with the same medium that supports topological defects, irreversible wakes, and gravitational gradients. The difference from the de Broglie–Bohm pilot-wave interpretation is that SSV does not introduce a guiding wave on top of classical trajectories; the order parameter IS the physical medium in which all dynamics occurs.

6.3 Testable consequences

Two near-term checks follow from the reconnection model.

Q1. Reconnection signatures in controlled measurement systems. If detector clicks are accompanied by irreversible medium events, then sufficiently sensitive optomechanical or superconducting measurement platforms should show excess correlated acoustic, thermal, or phase-noise signatures at the transition from reversible coupling to recorded outcome.

Q2. Born-rule basin scaling. A reduced reconnection-threshold model should test whether branch basin volumes scale as $|c_k|^2$ for simple two-channel apparatus couplings. This is the calculation needed before the measurement section can be promoted from physical interpretation to formal derivation.

6.4 Open problems

1. *Bound-state spectra.* Deriving the hydrogen-like spectrum directly from the LogSE, without treating V_{eff} as an external input, is part of the Paper I programme.

2. *Precise uncertainty prefactor.* The 1/2 in (14) should emerge from the LogSE coherent-state minimum; the calculation is deferred to the Paper I programme.

3. *Multi-particle states.* The present treatment addresses single-order-parameter modes. Multi-particle states and exchange statistics require a field-theoretic extension of the Madelung picture, not developed here.

4. *Spin-1/2 and Fermi statistics.* The order parameter Ψ is a scalar; half-integer spin and fermion anti-symmetry require an extension to spinor order parameters or a separate topological argument. This is noted as a gap, not filled here.

7 Conclusion

SSV VII-a isolates the quantum-mechanics limit from the geometry and cosmogony claims elsewhere in the series. The paper's formal core is the Madelung bridge and the Schrödinger limit. Its interpretive core has three parts:

1. Quantisation reflects phase topology: discrete winding numbers follow from the single-valuedness of the order parameter, not from a separate quantisation axiom.

2. Uncertainty reflects hydraulic localisation stiffness: the quantum pressure (6) penalises tight localisation, recovering the Heisenberg bound from the equation of state.

3. Measurement reflects irreversible reconnection or wake-writing in the medium, sourcing irreversibility from topology change rather than from an external classical realm.

That is sufficient to make the paper standalone without requiring it to carry the Einstein-equation or cosmogony arguments.

References

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